



Portfolio Risk Measurement & VaR Analysis

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A Static View of Individual Risk

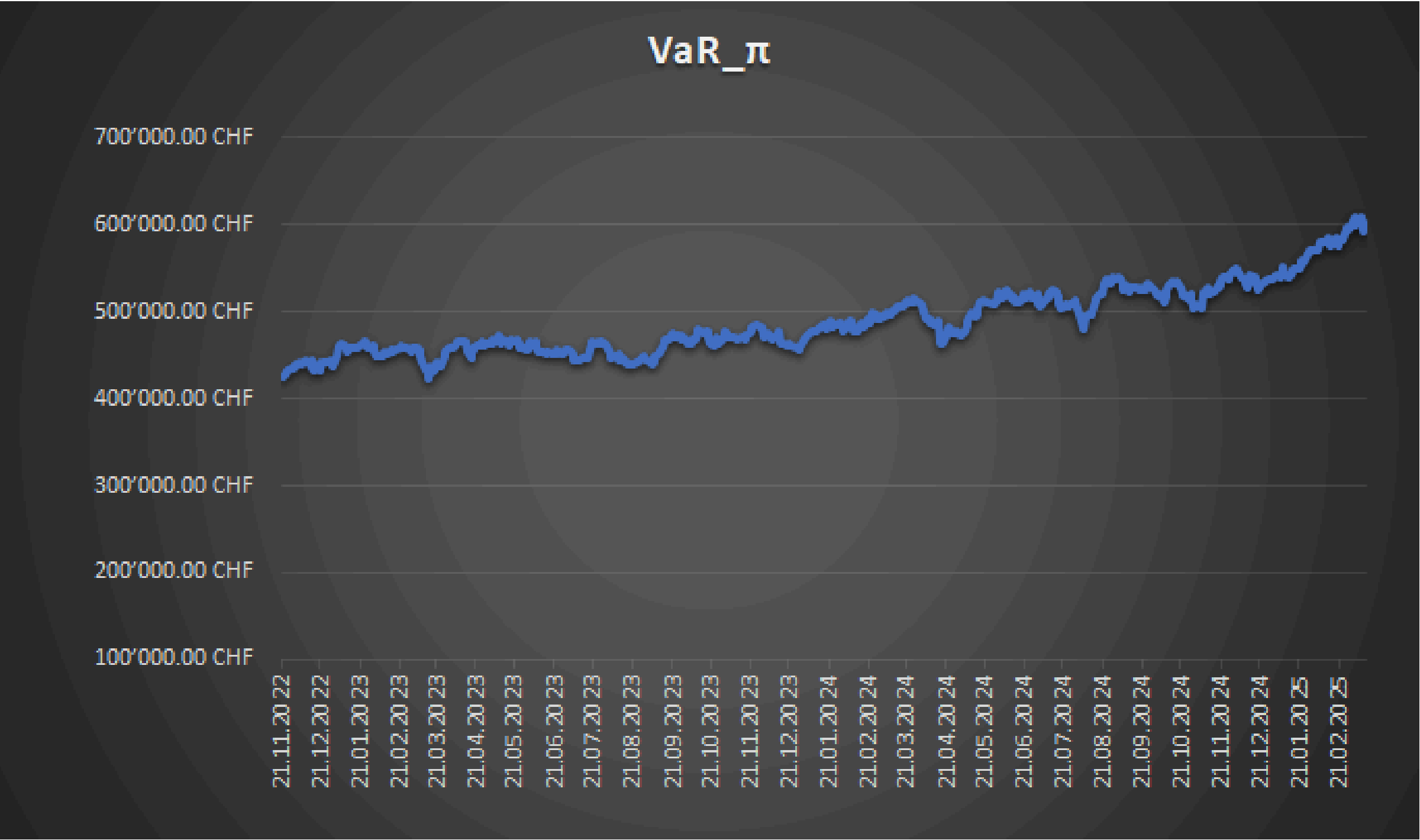


VaR(99%, 1-day) – Approach:

- Based on normal distribution and constant volatility
- Simple and fast to compute, but with limitations
- 250-day window = stable estimate, slower to adapt
- 125-day window = more responsive to recent changes
- May underestimate extreme (tail) risks

Stock	Volatility	VaR(99%,1d)	Position(shares)
NESN	1.09%	CHF 85,323	-30,000 (short)
SREN	1.47%	CHF 191,707	+70,000
NOVN	1.01%	CHF 147,486	+80,000

A Static View of Individual Risk

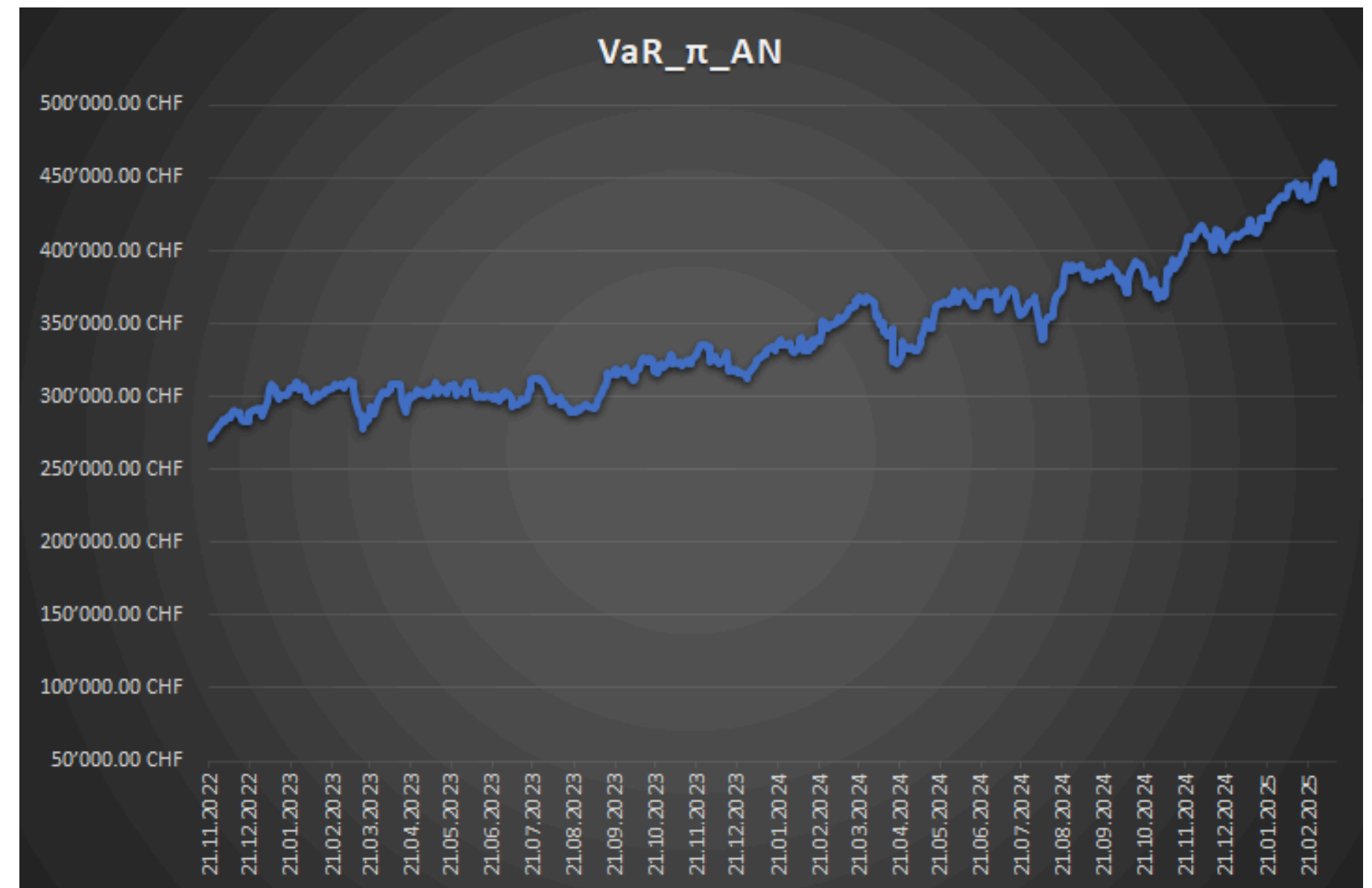


Diversification in Action



Portfolio VaR(99%, 1-day) – Asset-Normal Approach:

- Based on 250-day volatilities and correlations
- Buy-and-hold strategy (as of Nov 21, 2022)
- Assumes joint normality of asset returns
- Diversification reduces total risk
- Portfolio VaR < Sum of individual asset VaRs



Backtesting the Asset-Normal VaR



Backtesting VaR(99%, 1-day) – Asset-Normal Approach:

- Compared actual daily PnL vs. model VaR
- Exceedance: when $PnL < -VaR$
- Backtested on full sample and 2024 only
- Results not aligned with expectations
- 2024: higher exceedance rate → model underreacted to volatility

	P(VaR > PnL)	Exceedance Rate	Expected
Full Sample	98.40%	1.60%	1.00%
2024 Only	98.28%	1.72%	1.00%

RCVaR over Time



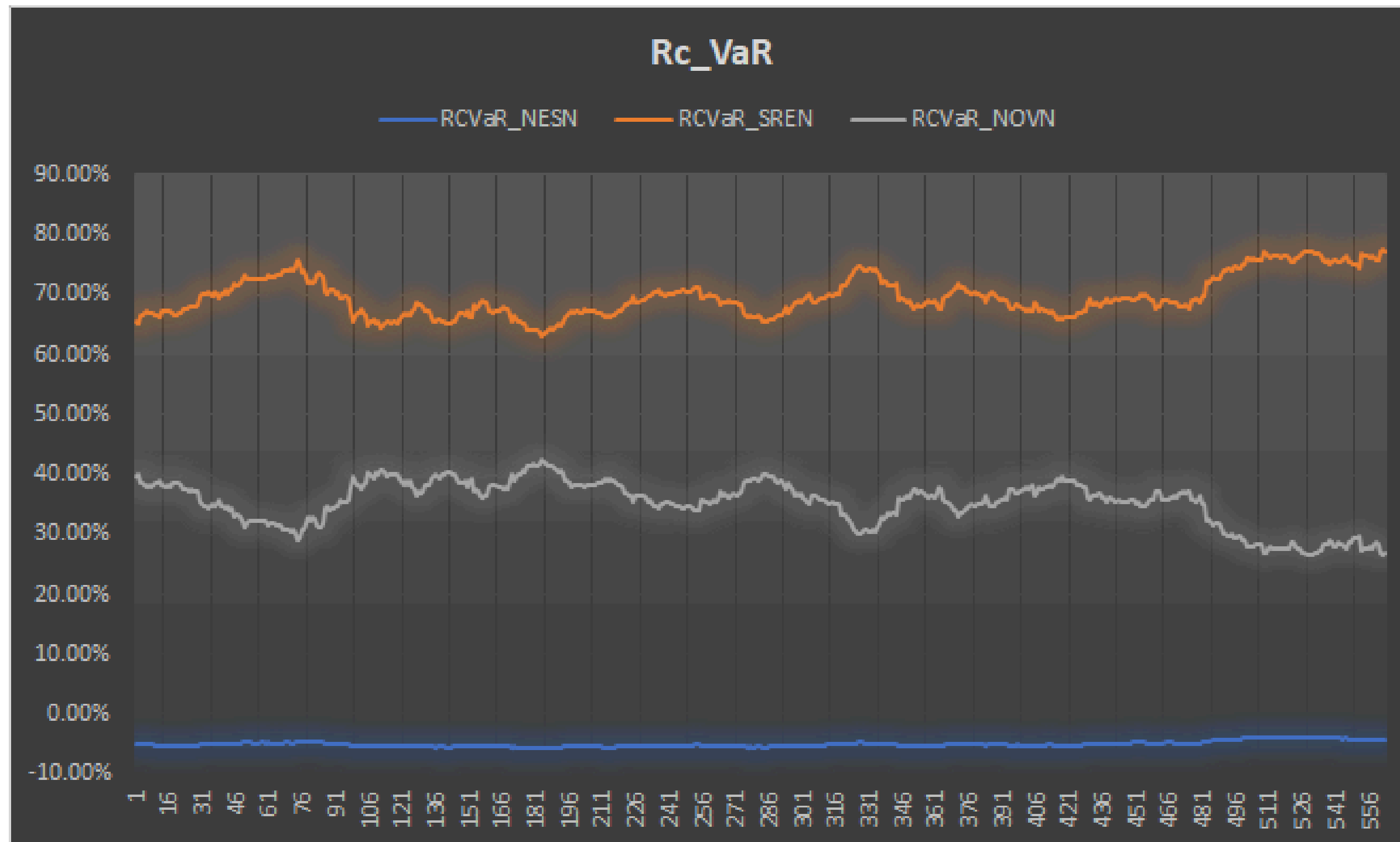
Risk Contribution Analysis:

- *MPaR: sensitivity of portfolio VaR to +1 CHF in a position*
- *CPaR = MPaR × position size*
- *RCVaR = CPaR / portfolio VaR*
- *Tracked RCVaRs over time to monitor each asset's risk impact*

Key Insights:

- *NESN (short): negative RCVaR → reduces total risk, especially in stressed markets*
- *SREN: largest contributor to VaR → high volatility & big position*
- *NOVN: lowest and most stable RCVaR → minor impact on risk*
- *RCVaR fluctuations driven by changes in volatility, correlation, and position size*

RCVaR over Time



Trading Impact on Portfolio Risk



Planned Trade – March 14, 2025:

- *Buy CHF 2M NESN*
- *Buy CHF 1M SREN*
- *Sell CHF 1M NOVN*

Metric	Value (CHF)
VaR Before Trades	454,128
VaR After Trades	489,436
Incremental VaR	+35,308

VaR Impact (Asset-Normal Model):

- *Initial estimate (IVaR) overstates risk – ignores diversification and covariances*
- *Full VaR recomputation shows only a modest risk increase*
- *Diversification benefits limit risk growth despite larger positions*
- *Final VaR confirms portfolio remains stable → trade is risk-efficient*

Systematic Risk via Market Betas



Beta Analysis – Swiss Market Index (SMI):

- *Estimated each stock's beta (β) using full return sample*
- *Beta = measure of systematic risk (co-movement with market)*
- *SREN: highest beta $\rightarrow$ more sensitive to market moves*
- *NESN: lowest beta $\rightarrow$ defensive stock characteristics*
- *Explains differences in Marginal VaR and shock sensitivity*

Stock	Beta (β)	Interpretation
NESN	0.757	Least market-sensitive
SREN	1.114	Most exposed to market risk
NOVN	0.901	Moderate exposure

EWMA vs. GARCH Volatility Modeling



Modeling Time-Varying Volatility of SMI Returns (Nov 2021 – Mar 2025):

EWMA ($\lambda = 0.94$) – RiskMetrics-style

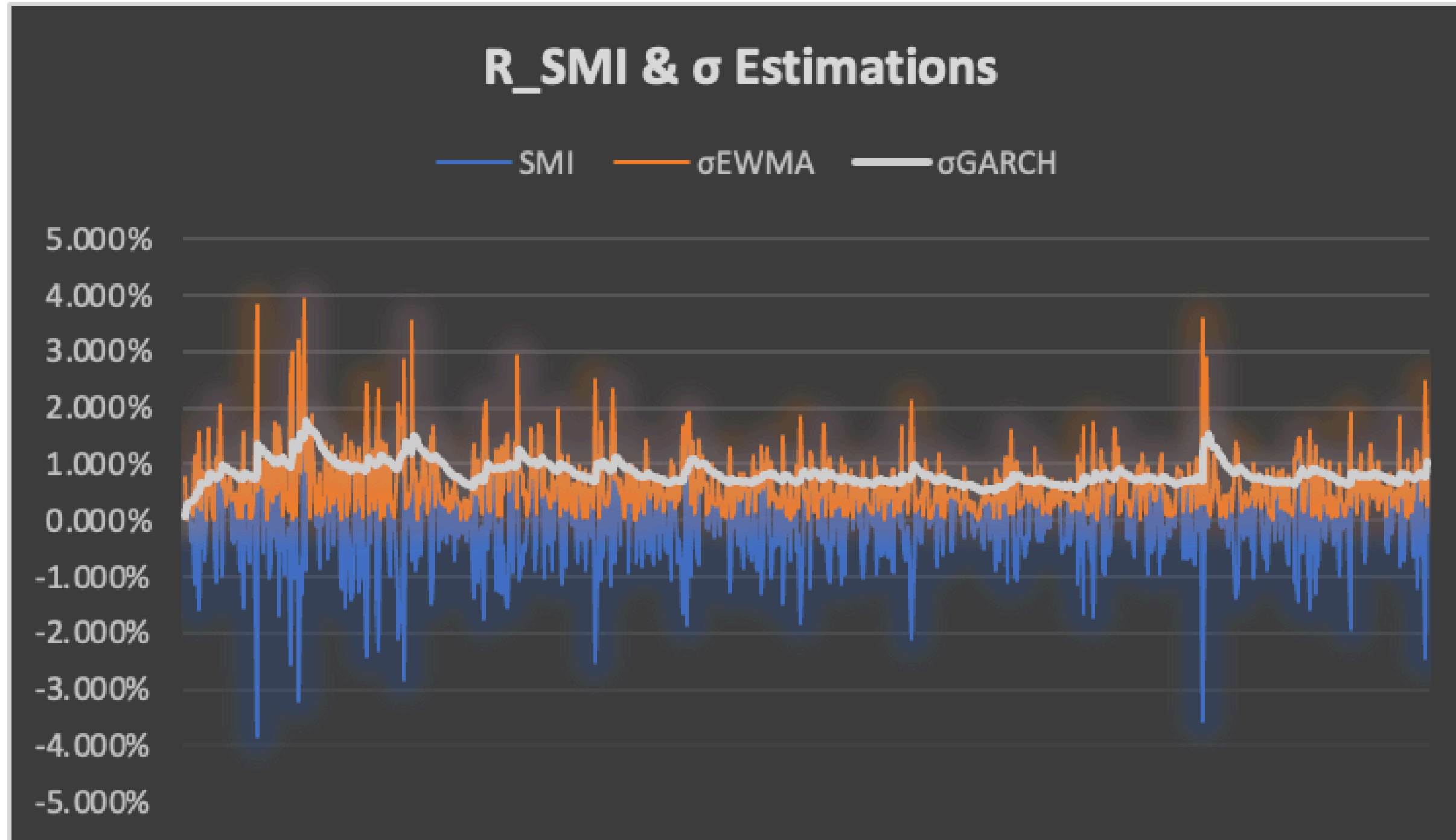
- *Smooth volatility estimates*
- *Reacts slowly to market shocks*
- *Assumes exponential decay of past shocks*

GARCH(1,1) – Volatility Clustering Model

- *Captures mean reversion and clustering*
- *More reactive to large return spikes*
- *Allows for persistent, autoregressive volatility*

Estimation	<i>EWMA</i>	GARCH
14/03/2025	0.36%	0.76%

EWMA vs. GARCH Volatility Modeling



Sharpe Diagonal VaR (99%, 1-day)



Compared to Asset-Normal VaR from Task 2 and based on Beta and Market Volatility

EWMA ($\lambda = 0.94$) Results:

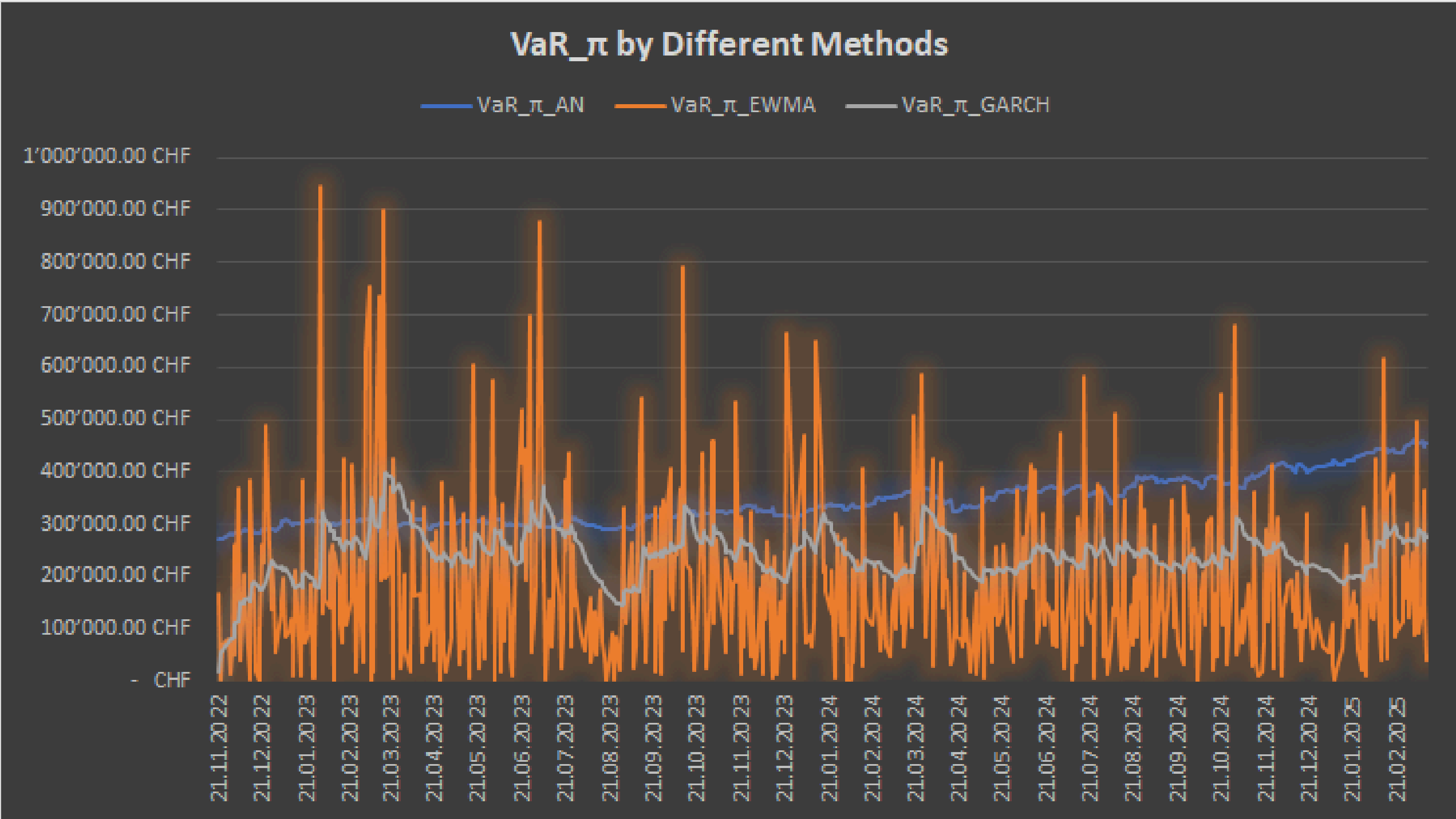
- *Underestimates portfolio risk*
- *Adjusts slowly, lacks mean reversion*
- *Less aligned with Asset-Normal VaR*

GARCH (1,1) Results:

- *Captures volatility clustering*
- *Reacts to market dynamics*
- *Closely matches Asset-Normal VaR*

Model	VaR (CHF)
EWMA	41,400
GARCH	276,209
Asset-Normal	454,128

Sharpe Diagonal VaR (99%, 1-day)





**Thank you for your
attention!**

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